
THE k -TH POWER EXPECTILE PERIODOGRAM FOR TIME SERIES ANALYSIS

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ABSTRACT

This paper introduces the k -th power expectile periodogram (KEP) as a generalization of the quantile periodogram (QP) and expectile periodogram (EP) for time series analysis. Constructed from trigonometric k -th power expectile regression (KER), the KEP provides a flexible tool to balance robustness and efficiency. The KEP retains the key properties of the ordinary periodogram as a frequency-domain representation of serial dependence in time series, while offering a more comprehensive understanding by examining the data across the entire range of expectile levels with power $1 < k \leq 2$. Simulations demonstrate the capability of the KEP in detecting hidden periodicities, while maintaining robustness against outliers. We establish the asymptotic theory and investigate the relationship between the KEP and the so-called k -th power expectile spectrum (KES). Finally, we leverage the inherent two-dimensional property of the KEP to train a convolutional neural network (CNN) to classify the epilepsy electroencephalogram (EEG) data, where the proposed estimator outperforms competitive estimators.

Keywords Expectile regression, Hidden periodicity, Periodogram, Time series analysis, Spectral analysis

1 Introduction

The ordinary periodogram (PG), a raw non-parametric estimator of the power spectrum, plays a crucial role in spectral analysis by identifying periodic components and frequency-domain structures. The PG is widely applied across various fields (Caiado et al., 2006; Polat and Güneş, 2007; Baud et al., 2018; Euán et al., 2018; Maadooliat et al., 2018; Martínez-Murcia et al., 2019; Chen et al., 2021a). Beyond the Fourier transform, the PG can also be obtained by ordinary least squares (OLS) trigonometric regression. The OLS regression primarily targets second-order properties

and is optimal under Gaussianity and linearity assumptions. However, real-world time series, especially in fields such as economics, medical research, and environmental sciences, often exhibit skewness, heavy tails, and nonlinear dependencies that violate these assumptions. Moreover, OLS regression focuses on the conditional mean, lacking the ability to provide a complete picture of the distribution of the data. Additionally, OLS regression is sensitive to outliers due to its quadratic loss function, which subsequently affects the performance of the PG (Bloomfield, 2004). As a result, quantile regression (QR) and expectile regression (ER) have been proposed to address the limitations of OLS regression.

The pioneering work of Koenker and Bassett Jr (1978) introduced the concept of QR, which has been comprehensively extended and studied Kouretas et al. (2005); Cai and Xu (2008); Cai and Xiao (2012); Koenker (2017). Newey and Powell (1987) proposed asymmetric least squares (ALS) regression, known as the ER, by substituting the ℓ_1 norm with the ℓ_2 norm in the check (loss) function of QR. Because of their significant advantages, QR and ER, as well as their extensions, have been widely applied across various scientific fields (Garcia et al., 2001; Machado and Mata, 2005; Gu and Zou, 2016; Ziegel, 2016; Bellini and Di Bernardino, 2017; Jiang et al., 2017; Daouia et al., 2018; Xu et al., 2020; Alvarado et al., 2021; Sharif et al., 2021; Xu et al., 2021).

Equipped with these regression techniques, new types of periodograms have been developed as counterparts to the PG in spectral analysis. Li (2012b) introduced the QP, which demonstrates the ability to detect hidden periodicities in time series. Large-sample theory showed that, like the properties of the power spectrum and the PG, the QP is an unbiased estimator of the quantile spectrum, which is a scaled version of the ordinary power spectrum of the level-crossing process. The Laplace periodogram (LP) (Li, 2008) represents a special case when the quantile level is set to 0.5. Chen (2024) further proposed the EP and comprehensively compared it with the QP. We refer to Katkovnik (1998); Liang et al. (2009); Li (2010, 2012a); Hagemann (2013); Li (2014); Kley (2014); Dette et al. (2015); Kley (2016); Birr et al. (2017); Meziani et al. (2020); Chen et al. (2021b); Li (2023) for related works. For more robust periodograms, Wen et al. (2021) proposed a Huber-loss based periodogram and Thieler et al. (2016) developed the R package RobPer, which allows users to combine different regression techniques (Rousseeuw and Yohai, 1984; Oh et al., 2004; Ahdesmäki et al., 2007; Thieler, 2014) to calculate robust periodograms.

The check function used in least absolute deviation ensures the robustness of QR, while the squared check function makes ER more sensitive to the tails of distributions and more efficient in the normal case. To balance robustness and effectiveness and make the regression more flexible in different situations, Jiang et al. (2021) introduced the KER with power $1 < k \leq 2$, as a generalization of both the QR and ER. Inspired by the KER and the achievements of the QP and EP, we propose a novel spectral estimator termed the KEP for spectral analysis of time series data. We demonstrate that the KEP not only shares similar properties with the PG as a frequency-domain representation of serial dependence in time series at a fixed expectile level, but also provides a more comprehensive analysis by exploring the full range of expectile levels and different values of the power k .

The remainder of the paper is organized as follows. In Section 2, we define the KEP and demonstrate its capability in spectral analysis through two real-world examples. The large sample theory reveals the asymptotic distribution of the

KEP, which is presented in Section 3. In Section 4, we present comparative studies evaluating the capability of the KEP in detecting hidden periodicities and its robustness against outliers through simulations. In Section 5, we leverage the inherent two-dimensional property of the KEP and train a CNN to classify the EEG data. Conclusions and future work are discussed in Section 6.

2 k -th Power Expectile Periodogram

2.1 k -th Power Expectile Regression

Suppose that $\{y_t\}$ is a stationary process with distribution function $F(\cdot)$. For any fixed $\alpha \in (0, 1)$ and $k \in (1, 2]$, define a loss function $\rho_{\alpha,k}(u) := |u|^k |\alpha - I(u < 0)|$, i.e.,

$$\rho_{\alpha,k}(u) := \begin{cases} \alpha |u|^k & \text{if } u \geq 0, \\ (1 - \alpha) |u|^k & \text{if } u < 0. \end{cases}$$

For $k = 1$ and $k = 2$, these loss functions are those used in QR (Koenker and Bassett Jr, 1978) and in ER (Newey and Powell, 1987), respectively. The k -th power α -expectile of y_t is then defined as

$$\mu(\alpha, k) := \arg \min_{\mu \in \mathbb{R}} \mathbb{E} \{ \rho_{\alpha,k}(y_t - \mu) \},$$

with the normal equation

$$\mathbb{E} \{ \dot{\rho}_{\alpha,k}(y_t - \mu(\alpha, k)) \} = 0. \quad (1)$$

Given a time series $\{y_t : t = 1, \dots, n\}$ and regressors $\{\mathbf{x}_{nt} : t = 1, \dots, n\}$ with $\mathbf{x}_{nt} := [x_{nt}(1), \dots, x_{nt}(p)]^T$ and $x_{nt}(1) := 1$, the KER at level α solves the optimization problem:

$$\hat{\beta}_n(\alpha, k) := \arg \min_{\beta \in \mathbb{R}^p} \sum_{t=1}^n \rho_{\alpha,k}(y_t - \mathbf{x}_{nt}^T \beta). \quad (2)$$

In the special case of $\alpha = 0.5, k = 2$, the KER problem (2) reduces to OLS regression because $\rho_{0.5,2}(u) = 0.5u^2$ for all $u \in \mathbb{R}$, and in the special case of $p = 1$, the KER solution $\hat{\beta}_n(\alpha, k) := \arg \min_{\beta \in \mathbb{R}} \sum_{t=1}^n \rho_{\alpha,k}(y_t - \beta)$ is called the sample k -th power α -expectile and is a natural estimator of $\mu(\alpha, k)$.

2.2 k -th Power Expectile Periodogram

Let $\omega_\nu := 2\pi\nu/n$ for $\nu = 0, 1, \dots, n-1$ denote the Fourier frequency for a time series of length n . Then, the regressor and the resulting KER solution are denoted by

$$\mathbf{x}_{nt}(\pi) := \begin{cases} [1, \cos(\omega_\nu t), \sin(\omega_\nu t)]^T & \text{if } \omega_\nu \notin \{0, \pi\}, \\ [1, \cos(\pi t)]^T & \text{if } \omega_\nu = \pi, \\ 1 & \text{if } \omega_\nu = 0, \end{cases} \quad (3)$$

and

$$\widehat{\beta}_n(\omega_\nu, \alpha, k) := \begin{cases} [\widehat{\beta}_1(\omega_\nu, \alpha, k), \widehat{\beta}_2(\omega_\nu, \alpha, k), \widehat{\beta}_3(\omega_\nu, \alpha, k)]^\top & \text{if } \omega_\nu \notin \{0, \pi\}, \\ [\widehat{\beta}_1(\pi, \alpha, k), \widehat{\beta}_2(\pi, \alpha, k)]^\top & \text{if } \omega_\nu = \pi, \\ \widehat{\beta}_1(0, \alpha, k) & \text{if } \omega_\nu = 0, \end{cases}$$

respectively.

Given these trigonometric KER solutions, the k -th power expectile discrete Fourier transform (KEDFT) of $\{y_t : t = 1, \dots, n\}$ at level α is defined as

$$z_n(\omega_\nu, \alpha, k) := \begin{cases} n\widehat{\beta}_1(0, \alpha, k) & \omega_\nu = 0, \\ n\widehat{\beta}_2(\pi, \alpha, k) & \omega_\nu = \pi \text{ (if } n \text{ is even)}, \\ (n/2)\{\widehat{\beta}_2(\omega_\nu, \alpha, k) - i\widehat{\beta}_3(\omega_\nu, \alpha, k)\} & \text{otherwise,} \end{cases}$$

where $i := \sqrt{-1}$. The KEDFT can be viewed as an extension of the ordinary DFT, which is defined by

$$z_n(\omega_\nu) := \sum_{t=1}^n y_t \exp(-i\omega_\nu t).$$

When $\alpha = 0.5$ and $k = 2$, the KEDFT coincides with the ordinary DFT, i.e., $z_n(\omega_\nu, 0.5, 2) = z_n(\omega_\nu)$ for all $\nu = 0, 1, \dots, n-1$. Then, the KEP at expectile level α and power k is defined as

$$\text{KEP}_n(\omega_\nu, \alpha, k) := n^{-1}|z_n(\omega_\nu, \alpha, k)|^2. \quad (4)$$

Below are the special cases of the KEP:

- Ordinary periodogram, when $\alpha = 0.5, k = 2$.
- Expectile periodogram (Chen, 2024), when $k = 2$.
- Laplace periodogram (Li, 2008), when $\alpha = 0.5, k = 1$.
- Quantile periodogram (Li, 2012b), when $k = 1$.

Remarks:

- Efron (1991) and Jiang et al. (2021) pointed out that the power loss function with $k = 1.5$ is appealing as a compromise between the robustness of $k = 1$ and the high-normal-theory efficiency of $k = 2$. In contrast to QR, the common ER requires that the mean of the underlying distribution exists, which can be restrictive, especially for financial data. The KER relaxes this requirement to the existence of the $(k-1)$ -th order moment. QR requires no moment condition, but the computation involved, for instance in calculating the variance, is not so easy as that of the KER. As a result, we restrict $1 < k \leq 2$ in this paper. For $k = 1$, we refer to Li (2012b).
- It is convenient to define $\text{KEP}_n(0, \alpha, k) = 0$ instead of $n|\widehat{\beta}_1(0, \alpha, k)|^2$. This is analogous to the property of the PG of the demeaned series $\{y_t - \bar{y}_t : t = 1, \dots, n\}$.
- In Section 4 and 5, we emphasize on serial dependence of the time series, with amplitude being secondary importance, so we normalize all the periodograms at each expectile level such that the summation over ω

equals 1. Additionally, due to symmetry, we only consider restricted $\omega \in [0, \pi]$, and use $f = \omega/2\pi \in [0, 1/2]$, which represents the number of cycles per unit time, in all figures in this paper.

2.3 Examples

When k is fixed, the KEP can be analyzed at a specific α like the PG, it can also be treated as a two-dimensional estimator of parameters α and ω . To illustrate such duality, we consider two examples in different scientific fields, in which the PG fails to capture the full spectral features of the time series.

In the first example, we analyze the daily log returns of the S&P 500 Index (1986 – 2015). As shown in Figure 1, the KEPs in the second and third rows successfully identify the approximately 10-year cycle of market volatility at both lower and upper expectiles. In contrast, the PG yields a relatively featureless flat line. The figures in the last row demonstrate that as k increases, the KEPs across the expectile level at a fixed frequency become smoother. We smooth the KEP ($\alpha = 0.05, 0.07, \dots, 0.95$ and $k = 1.9$) at each expectile level and observe that the spectral feature is similar to the KES (see Section 3 for details) of a GARCH(1,1) model (Engle, 1982; Bollerslev, 1986):

$$y_t \sim N(0, \sigma_t^2), \quad (5)$$

where $\sigma_t^2 = 10^{-6} + 0.49y_{t-1}^2 + 0.49\sigma_{t-1}^2$, as shown in Figure 2. The KEP captures the spectral patterns of the GARCH(1,1) model at both lower and upper expectiles at low frequencies, while in ordinary spectral analysis, the power spectrum is a constant.

The second example is based on epilepsy electroencephalogram (EEG) data¹. As shown in Figure 3, the PG in Figure 3 (b) successfully detects both low-frequency and high-frequency components, corresponding to the six main spikes and the associated bursts, respectively. However, the KEPs provide additional information, showing that the high-frequency patterns appear predominantly at higher expectile levels. This observation is consistent with the EEG time series, where the bursts primarily occur at the top of the main spikes.

3 Asymptotic Analysis

In this section, we develop the large-sample asymptotic analysis, which reveals an association between the KEP and what we call the KES, which is analogous to the relationship between the PG and the power spectrum

For fixed $\alpha \in (0, 1)$ and $k \in (1, 2]$, let $\{y_t\}$ be a stationary process with cumulative distribution function $F(\cdot)$. Consider the KEP at q distinct frequencies, denoted by $\{\omega_{\nu_j} : j = 1, \dots, q\}$. Let $\mathbf{x}_{jnt} = [1, \cos(\omega_{\nu_j}t), \sin(\omega_{\nu_j}t)]^T$, $t = 1, \dots, n$, $j = 1, \dots, q$. Substituting $\mathbf{x}_{nt}$ with $\mathbf{x}_{jnt}$ in (2), we obtain $\hat{\beta}_{jn}(\alpha, k)$, and $\text{KEP}_n(\omega_{\nu_j}, \alpha, k)$. We define

$$\zeta_{jn}(\alpha, k) := n^{-1/2} \sum_{t=1}^n \dot{\rho}_{\alpha, k}(y_t - \mu(\alpha, k)) \mathbf{x}_{jnt}, \quad (6)$$

¹The data were collected by the School of Psychology at South China Normal University and authorized for use in this paper.

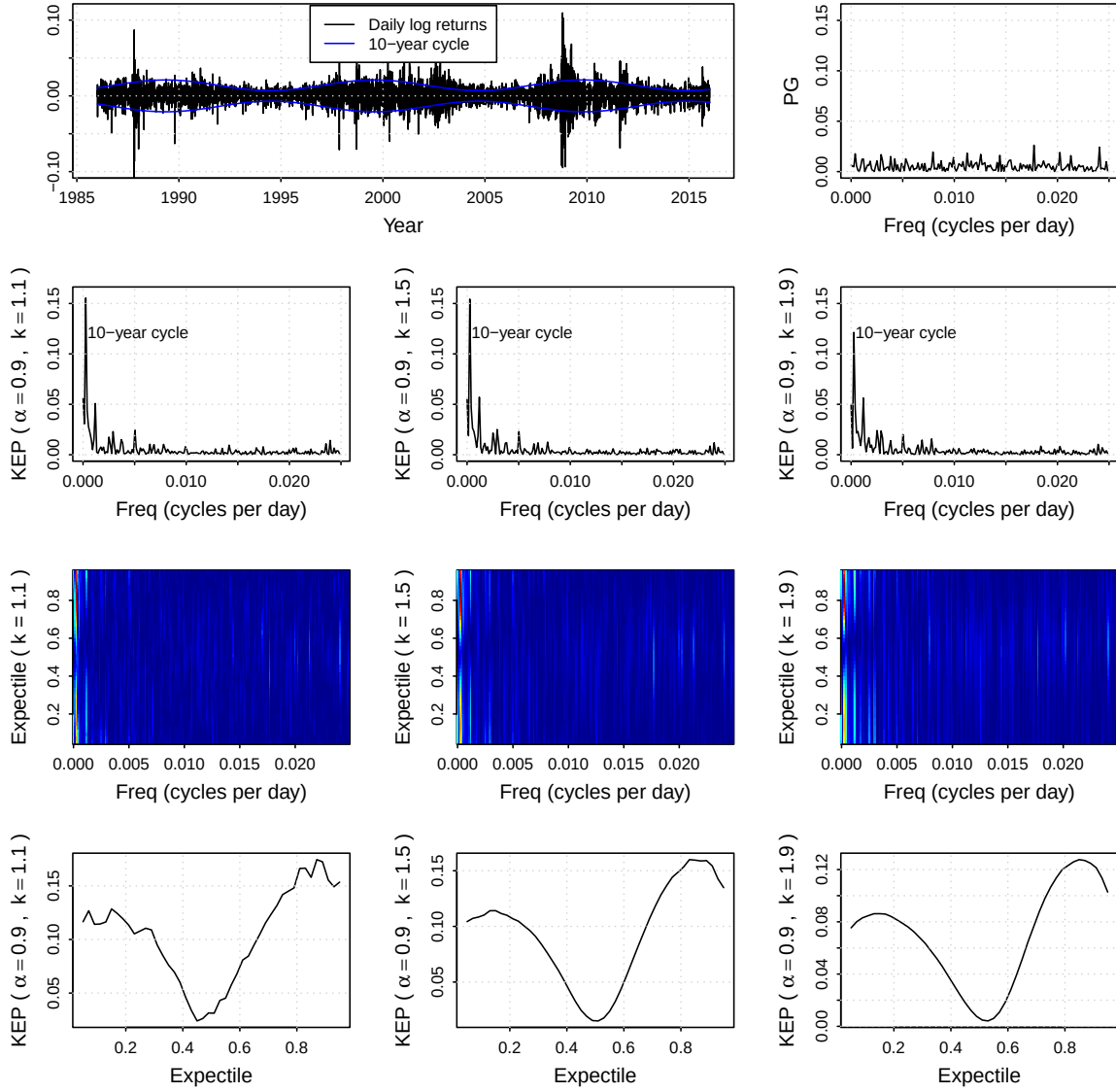


Figure 1: First row (left): daily log returns of the S&P 500 Index data, along with its 10-year cycle; first row (right): the PG. Second row: the KEPs at $\alpha = 0.9$ (left to right: $k = 1.1, 1.5, 1.9$). Third row: the KEPs at $\alpha = 0.05, 0.07, \dots, 0.95$ (left to right: $k = 1.1, 1.5, 1.9$). Fourth row: the KEPs across the expectile level at a fixed frequency $\omega = 2/7560 \times 2\pi$ (left to right: $k = 1.1, 1.5, 1.9$).

where

$$\dot{\rho}_{\alpha,k}(u) = \begin{cases} \alpha k u^{k-1} & \text{if } u \geq 0, \\ -(1-\alpha)k(-u)^{k-1} & \text{if } u < 0. \end{cases} \quad (7)$$

Let the following conditions be satisfied for fixed α and k :

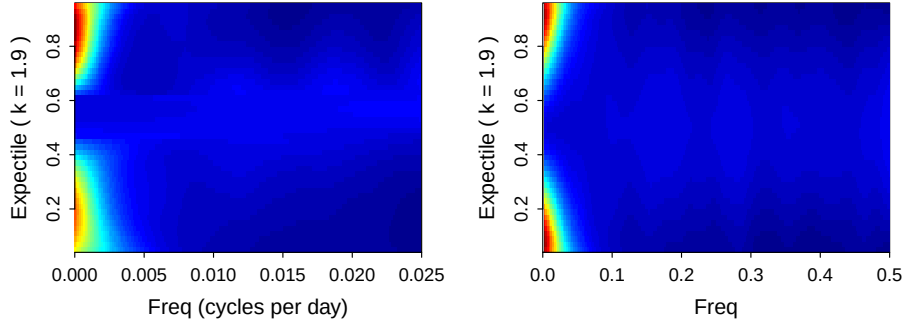


Figure 2: Left: the smoothed KEP ($\alpha = 0.05, 0.07, \dots, 0.95$ and $k = 1.9$). Right: the KES of model (5), obtained by averaging the smoothed KEPs from 2,000 realizations. The smoothing uses `smooth.spline` function in R.

C1. The cumulative distribution function $F(\cdot)$ is Lipschitz continuous, i.e., there exists a constant $K > 0$ such that

$$|F(y) - F(y')| \leq K|y - y'| \text{ for all } y, y' \in \mathbb{R}.$$

C2. The process $\{\dot{\rho}_{\alpha,k}(y_t - \mu(\alpha, k))\}$ is stationary with zero mean and absolutely summable ACF, defined as

$$\text{Cov}\{\dot{\rho}_{\alpha,k}(y_t - \mu(\alpha, k)), \dot{\rho}_{\alpha,k}(y_s - \mu(\alpha, k))\} = \gamma(t - s, \alpha, k).$$

C3. $E|y_t|^{k+2} < c_0$.

C4. $1.5 < k \leq 2$.

C5. A central limit theorem is valid for $\text{vec}[\zeta_{jn}(\alpha, k)]_{j=1}^q$ for all j , i.e., $\text{vec}[\zeta_{jn}(\alpha, k)]_{j=1}^q \xrightarrow{D} N(\mathbf{0}, [\mathbf{V}_{jl}(\alpha, k)]_{j,l=1}^q)$ as $n \rightarrow \infty$, where $\mathbf{V}_{jl}(\alpha, k)$ is the asymptotic covariance matrix of $\zeta_{jn}(\alpha, k)$ and $\zeta_{ln}(\alpha, k)$.²

By C2, we define

$$g(0, \alpha, k) := \eta^2(\alpha, k)h(0, \alpha, k), \text{ and } g(\omega, \alpha, k) := \eta^2(\alpha, k)h(\omega, \alpha, k), \quad (8)$$

where $h(0, \alpha, k) := \sum_{\tau} \gamma(\tau, \alpha, k)$, and

$$h(\omega, \alpha, k) := \sum_{\tau=-\infty}^{\infty} \gamma(\tau, \alpha, k) \exp(-i\omega\tau) = \sum_{\tau=-\infty}^{\infty} \gamma(\tau, \alpha, k) \cos(\omega\tau). \quad (9)$$

The second expression in (9) is due to the symmetry $\gamma(\tau, \alpha, k) = \gamma(-\tau, \alpha, k)$. We denote $U_t := y_t - \mu(\alpha, k)$. Suppose that $\Pr\{|U_t| > 0\} > 0$, which indicates that the distribution of $\{y_t\}$ is not degenerate. Then $E\{\ddot{\rho}_{\alpha,k}(U)\}$ is strictly nonzero, and the scaling factor $\eta(\alpha, k)$ is defined as

$$\begin{aligned} \eta^{-1}(\alpha, k) &= E\{\ddot{\rho}_{\alpha,k}(U)\} \\ &= k(k-1) [\alpha E\{|U|^{k-2}I(U > 0)\} + (1-\alpha) E\{|U|^{k-2}I(U < 0)\}] \\ &= k(k-1) \left[\alpha \int_0^{\infty} u^{k-2} dF(u + \mu(\alpha, k)) + (1-\alpha) \int_{-\infty}^0 (-u)^{k-2} dF(u + \mu(\alpha, k)) \right], \end{aligned}$$

²This is a relaxed condition, see Remark 1 in Li (2012b) and Condition C3 in Chen (2024) for details.

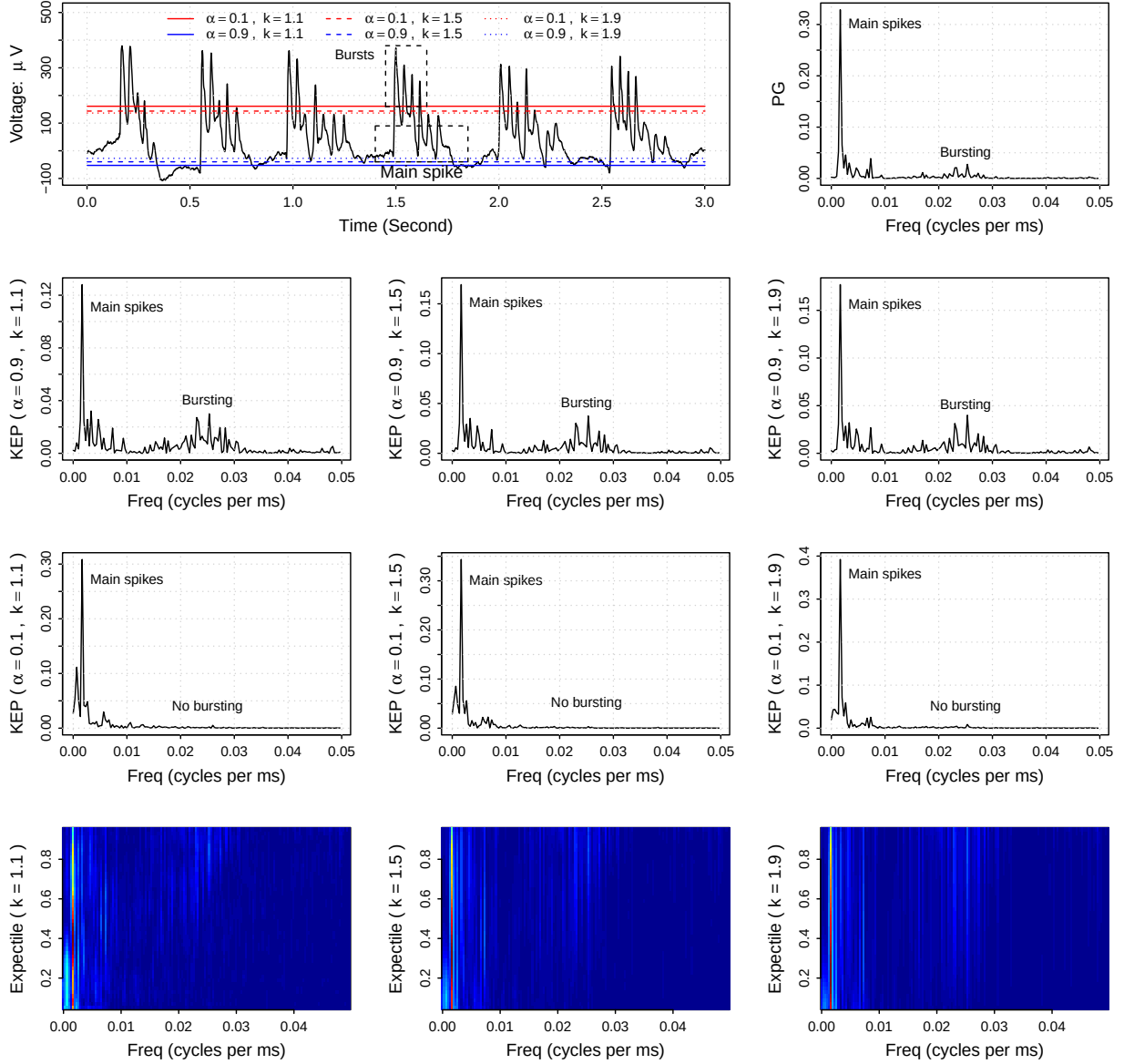


Figure 3: First row (left): the EEG data over a 3-second interval, along with its sample expectiles; first row (right): the PG. Second row: the KEPs at $\alpha = 0.9$ (left to right: $k = 1.1, 1.5, 1.9$). Third row: the KEPs at $\alpha = 0.1$ (left to right: $k = 1.1, 1.5, 1.9$). Fourth row: the KEPs at $\alpha = 0.05, 0.07, \dots, 0.95$ (left to right: $k = 1.1, 1.5, 1.9$).

which depends solely on the marginal distribution $F(\cdot)$, where

$$\ddot{\rho}_{\alpha,k}(u) = k(k-1) |\alpha - I\{u < 0\}| \cdot |u|^{k-2}. \quad (10)$$

Equipped with these concepts, we establish the following theorem regarding the $\text{KEP}_n(\omega_{\nu_j}, \alpha, k)$ in (4).

Theorem 1 (k -th power Expectile Periodogram). For fixed $q > 1$ and $0 < \lambda_1 < \dots < \lambda_q < \pi$, let $\omega_{\nu_1}, \dots, \omega_{\nu_q}$ be Fourier frequencies satisfying $\omega_{\nu_j} \rightarrow \lambda_j$ as $n \rightarrow \infty$ for $j = 1, \dots, q$. If (C1) - (C5) are satisfied, then

$$\{\text{KEP}_n(\omega_{\nu_1}, \alpha, k), \dots, \text{KEP}_n(\omega_{\nu_q}, \alpha, k)\} \xrightarrow{D} \{g(\lambda_1, \alpha, k)(1/2)\chi_{2,1}^2, \dots, g(\lambda_q, \alpha, k)(1/2)\chi_{2,q}^2\}, \quad (11)$$

where $\chi_{2,1}^2, \dots, \chi_{2,q}^2$ are independent standard chi-square random variables with two degrees of freedom.

Proof. See Appendix A.

Theorem 1 shows that the KEP exhibits a scaled chi-square distribution with two degrees of freedom, which is similar to the behavior of the PG (Brockwell and Davis, 1991). The definition of $h(\omega, \alpha, k)$ in (9) can be extended to include all $\omega \in [0, 2\pi)$, and it is essentially the ordinary spectrum of the stationary process $\{\dot{\rho}_{\alpha,k}(y_t - \mu(\alpha, k))\}$. This process is an asymmetrically-scaled version of the process $\{y_t - \mu(\alpha, k)\}$ around $\mu(\alpha, k)$. Similarly, as a scaled version of $h(\omega, \alpha, k)$, the definition of $g(\omega, \alpha, k)$ can also be extended to include all $\omega \in [0, 2\pi)$, which we define as the KES of $\{y_t\}$.

4 Simulation Studies

In this section, we present experiments to evaluate the effectiveness of the KEP in detecting hidden periodicities and its robustness against outliers.

4.1 Hidden Periodicities Detection

We consider two models, $\{x_t\}$ and $\{y_t\}$, representing models without and with hidden periodicities, respectively. Specifically, $\{x_t\}$ is an AR(2) model:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \epsilon_t, \quad (12)$$

and

$$y_t = a_t x_t, \quad (13)$$

where

$$a_t = b_0 + b_1 \cos(\omega_{\nu_0} t) + b_2 \sin(\omega_{\nu_1} t), \quad (14)$$

with $b_0 = 1, b_1 = 0.9, b_2 = 1, \phi_1 = 2r \cos(\omega_{\nu_c}), \phi_2 = -r^2$ ($r = 0.6$), and $\{\epsilon_t\}$ is standard Gaussian white noise. Additionally, we set $\omega_{\nu_0} = 0.09 \times 2\pi, \omega_{\nu_1} = 0.12 \times 2\pi$ as the frequencies of the hidden periodicities.

We first present the periodograms of model (12), representing the case without hidden periodicities. For comparison, we include the PG and LP, which focus only on the conditional mean and median, respectively. As can be seen in Figure 4 (a) and (b), where the power k is set to 1.5, all types of periodograms exhibit similar bell-shaped patterns, with spectral peaks around $\omega_{\nu_c} = 0.25 \times 2\pi$. Here, all the periodograms are obtained by averaging smoothed periodograms from 2,000 simulation realizations. These results can be directly explained by ordinary spectral analysis. Specifically, the characteristic polynomial of model (12) is $\phi(z) = 1 - \phi_1 z - \phi_2 z^2$, where the roots z_1 and z_2 are complex conjugates.

The magnitude $|z_1| = |z_2| = 1/r > 1$ ensures causality. The AR coefficients ϕ_1 and ϕ_2 determine the location of the spectral peak:

$$\omega = \arccos \frac{\phi_1}{2\sqrt{-\phi_2}} = \omega_{\nu_c},$$

and as $r \rightarrow 1^-$, the peak becomes narrower.

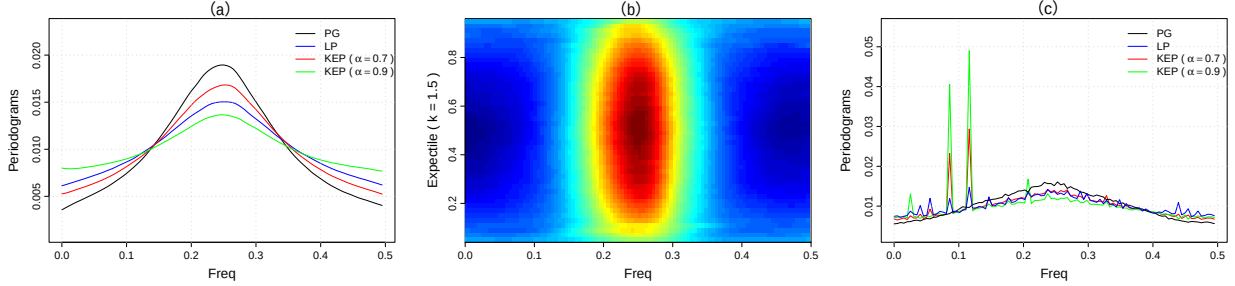


Figure 4: Left: averaged smoothed PG, LP, and KEP ($\alpha = 0.7, 0.9$ and $k = 1.5$) of model (12). Middle: averaged smoothed KEP ($\alpha = 0.05, 0.07, \dots, 0.95$ and $k = 1.5$) of model (3). Right: PG, LP, and KEP ($\alpha = 0.7, 0.9$ and $k = 1.5$) of model (13). The number of realizations is 2,000 with sample size $n = 200$. The smoothing uses `smooth.spline` function in R.

In the case of model (13), as shown in Figure 4(c), the KEPs successfully detect the hidden periodicities, manifesting large spikes at $\omega_{\nu_0}/2\pi = 0.09$ and $\omega_{\nu_1}/2\pi = 0.12$, whereas the PG and LP do not exhibit this capability, instead yielding similar bell-shaped patterns as in model (12). The results are obtained by averaging periodograms from 2,000 simulation runs. We observe that the spectral leakage, which appears in the LP, can also occur in the KEP. Therefore, small spikes may take place at some other frequencies, as indicated by Theorem 5 in Li (2008) and Theorem 2 in Li (2012b). To address this issue, incorporating LASSO regularization into the loss function is beneficial (Meziani et al., 2020).

The KEPs in Figure 2 and Figure 4(b) are symmetric across the expectile level. To demonstrate the capability of the KEP in handling more complex spectral features, we construct a nonlinear mixture model $\{y_t\}$, whose KEP is intentionally made asymmetric across the expectile level:

$$\begin{aligned} z_t &= W_1(x_{t,1}) x_{t,1} + \{1 - W_1(x_{t,1})\} x_{t,2}, \\ y_t &= W_2(z_t) z_t + \{1 - W_2(z_t)\} x_{t,3}. \end{aligned} \tag{15}$$

The components $\{x_{t,1}\}$, $\{x_{t,2}\}$, and $\{x_{t,3}\}$ are independent Gaussian AR processes given by

$$\begin{aligned} x_{t,1} &= 0.8x_{t-1,1} + w_{t,1}, \\ x_{t,2} &= -0.75x_{t-1,2} + w_{t,2}, \\ x_{t,3} &= -0.81x_{t-2,3} + w_{t,3}, \end{aligned}$$

where $w_{t,1}, w_{t,2}, w_{t,3}$ are standard Gaussian white noise. From the perspective of traditional spectral analysis, the series $\{x_{t,1}\}$ has a lowpass spectrum, $\{x_{t,2}\}$ has a highpass spectrum, and $\{x_{t,3}\}$ has a bandpass spectrum around frequency 0.25. The mixing function $W_1(x)$ is equal to 0.9 for $x < -0.8$, 0.2 for $x > 0.8$, and a linear transition for x in between. The mixing function $W_2(x)$ is similarly defined but equals 0.5 for $x < -0.4$ and 1 for $x > 0$. Figure 5 shows the averaged smoothed PG and KEPs of model (15) from 2,000 simulations, where the KEP is asymmetric across the expectile levels. We also compare the PG and the KEPs at $\alpha = 0.1$ and 0.9, illustrating that the PG is restricted to capturing spectral features near the central expectiles. In contrast, the KEP offers a broader perspective, effectively analyzing the time series across the entire range of $\alpha \in (0, 1)$.

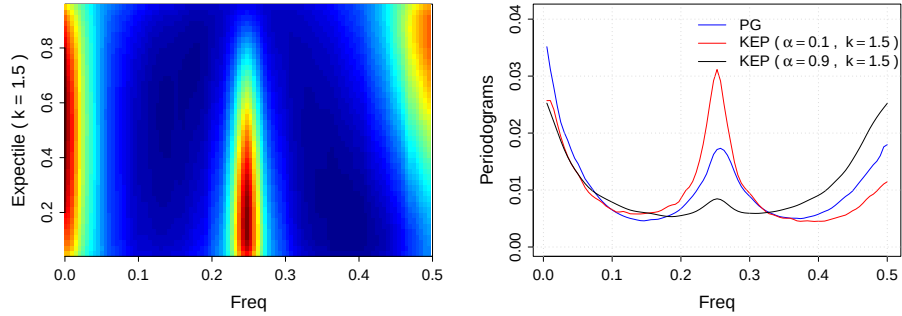


Figure 5: Left: KEP ($\alpha = 0.05, 0.07, \dots, 0.95$ and $k = 1.5$). Right: PG and KEP ($\alpha = 0.1, 0.9$ and $k = 1.5$). We use the average of 2,000 smoothed periodograms to generate the results with sample size $n = 200$. The smoothing uses `smooth.splines` function in R.

4.2 Robustness Evaluation using Fisher's Test

One commonly used hypothesis test to detect periodicities for the PG is Fisher's test (Brockwell and Davis, 1991). For frequencies $\{\omega_{\nu_1}, \omega_{\nu_2}, \dots, \omega_{\nu_q}\}$, the test statistic is defined as

$$g = \frac{\max_{1 \leq j \leq q} \{I_n(\omega_{\nu_j})\}}{\sum_{j=1}^q I_n(\omega_{\nu_j})}.$$

According to Fisher's test, a sufficiently large value of g indicates the presence of a hidden periodicity. The null hypothesis is that the time series is Gaussian white noise, while the alternative hypothesis is that the time series contains a deterministic periodic component of an unspecified frequency. We apply Fisher's test to the KEP by replacing $I_n(\omega)$ with $\text{KEP}_n(\omega, \alpha, k)$ in the test statistic.

We consider the model

$$y_t = a_t + x_t, \tag{16}$$

where x_t and a_t are defined by (12) and (14), respectively. Moreover, b_1 is set to 0, indicating that only a single periodicity is present. To evaluate the robustness of the KEP, we contaminate model (16) by adding a constant c to $\{y_t\}$

at a single time point t^* :

$$y_t^* = \begin{cases} y_t & \text{if } t \neq t^* \\ y_t + c & \text{if } t = t^*, \end{cases} \quad (17)$$

where t^* is a random integer in $[1, n]$. The left column in Figure 6 depicts a realization of $\{y_t\}$ and $\{y_t^*\}$, respectively.

We generate 2,000 time series based on models (16) and (17), respectively, and compute their PG and KEP ($k = 1.1, 1.2, 1.3$ and $\alpha = 0.5, 0.9$). Then, we compute the power of Fisher’s test and obtain the probability of detection (PD) under three significance levels of 0.01, 0.05, 0.1 with $c = 50, 60, 70$. Table 1 lists the PD, along with their differences (PD of y_t minus PD of y_t^*). The results show that periodicity can be reliably detected by both PG and KEP in the absence of outliers, with all values in the top three rows close to 1. This can be further validated by the top row of Figure 6, in which both the PG and KEP exhibit large spikes at $\omega_{\nu_0} = 0.1 \times 2\pi$. However, when outliers are included, the PD for PG decreases rapidly, resulting in large differences. In comparison, the PD for the KEP remains more stable, resulting in smaller differences. The bottom row of Figure 6 also illustrates that the large peak at ω_{ν_0} , persists in the KEP but disappears in the PG. The bell-shaped pattern in the PG is noticeably diminished. On average, the differences for KEP are only 0.1001, 0.1101, 0.1191 when $c = 50, 60, 70$, respectively, compared to 0.9067, 0.9882, 0.9997 for PG. This also indicates that the difference becomes larger as c increases. We further observe that the robustness of the KEP improves as k approaches 1, though at the cost of reduced efficiency, as reflected by the average estimation times in Table 1. All computations were performed on a PC equipped with an Intel Core i9-13900KF CPU @5.2 GHz, 64 GB of RAM, and an Nvidia RTX 4090 GPU.

5 Epilepsy EEG Data Classification

5.1 Experiment Settings

In this application, we use the EEG segments described in Andrzejak et al. (2001) to classify subjects into different groups. There are five datasets (denoted as A, B, C, D, and E), collected from individuals in different states. Each set contains 100 single-channel EEG segments of length $n = 4,000$. The content of the five sets is:

- A: Healthy volunteers with eyes open;
- B: Healthy volunteers with eyes closed;
- C: Epileptic patients in seizure-free intervals from the epileptic hemisphere;
- D: Epileptic patients in seizure-free intervals from the opposite hemisphere;
- E: Epileptic patients during seizure intervals.

For each EEG recording, we split the long time series into 10 non-overlapping segments of length 400, yielding a total of 5,000 segments. We then consider four different classification tasks, following Orhan et al. (2011) and Maadooliat et al. (2018), which vary in difficulty, to evaluate the performance of the KEP in distinguishing mixtures of healthy segments, seizure-free epileptic segments, and seizure segments:

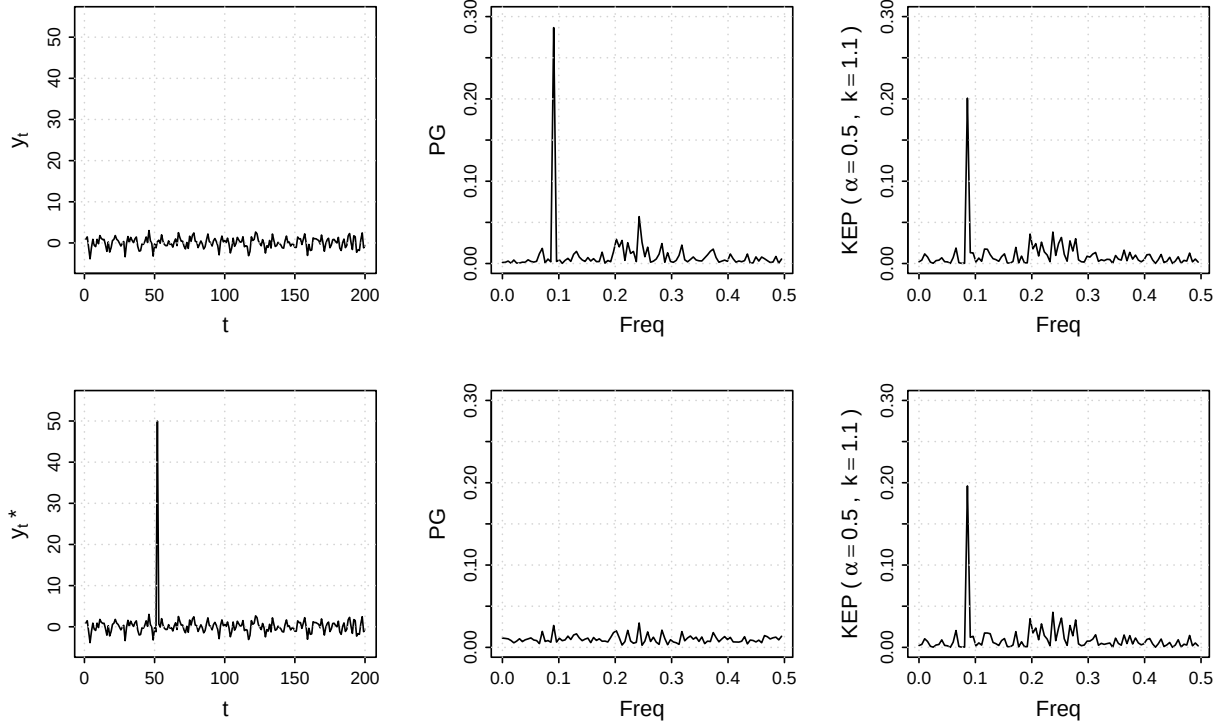


Figure 6: Top row: y_t with no outliers. Bottom row: y_t^* with outliers. Left column: time series. Middle column: PG. Right column: KEP ($\alpha = 0.5$ and $k = 1.1$). The amplitude of the outlier is $c = 50$.

- Easy Task (A vs. E): This task aims to distinguish between epileptic seizure patients (E) and healthy volunteers (A). Therefore, we classify 2,000 segments into 2 classes.
- Intermediate Task (ABCD vs. E): This task aims to distinguish between patients with seizure activity (E) and others (ABCD). Therefore, we classify 5,000 segments into 2 classes.
- Hard Task 1 (A vs. D vs. E): This task aims to distinguish EEG signals from healthy volunteers (A), seizure-free epileptic patients (D), and patients experiencing seizures (E). Therefore, we classify 3,000 segments into 3 classes.
- Hard Task 2 (AB vs. CD vs. E): This task aims to distinguish EEG signals from healthy volunteers (AB), epileptic patients in seizure-free intervals (CD), and patients experiencing seizures (E). Therefore, we classify 5,000 segments into 3 classes.

We compute the KEP of each segment with frequencies restricted to $\omega \leq 0.25 \times 2\pi$, expectile levels $\alpha = 0.05, 0.07, \dots, 0.95$, and power parameters $k = 1.1, 1.5$, and 1.9 . In addition, we use the PG as a benchmark method.

Table 1: Fisher’s test of different types of periodograms.

Model	Significa- nce level	KEP ($k = 1.1$)		KEP ($k = 1.2$)		KEP ($k = 1.3$)		PG
		$\alpha = 0.5$	$\alpha = 0.9$	$\alpha = 0.5$	$\alpha = 0.9$	$\alpha = 0.5$	$\alpha = 0.9$	
y_t	0.01	0.9980	0.7655	1.0000	0.8605	1.0000	0.9210	1.0000
	0.05	1.0000	0.8895	1.0000	0.9525	1.0000	0.9785	1.0000
	0.10	1.0000	0.9335	1.0000	0.9720	1.0000	0.9895	1.0000
$y_t^*(c = 50)$	0.01	0.9975	0.6255	0.9990	0.6080	0.9995	0.4785	0.0040
	0.05	0.9995	0.8025	1.0000	0.7985	1.0000	0.6675	0.0905
	0.10	1.0000	0.8680	1.0000	0.8600	1.0000	0.7545	0.1855
Diff. ($c = 50$)	0.01	0.0005	0.1400	0.0010	0.2525	0.0005	0.4425	0.9960
	0.05	0.0005	0.0870	0.0000	0.1540	0.0000	0.3110	0.9095
	0.10	0.0000	0.0655	0.0000	0.1120	0.0000	0.2350	0.8145
$y_t^*(c = 60)$	0.01	0.9975	0.6225	0.9990	0.5935	0.9995	0.4335	0.0000
	0.05	0.9995	0.7980	1.0000	0.7840	1.0000	0.6230	0.0040
	0.10	1.0000	0.8665	1.0000	0.8495	1.0000	0.7120	0.0315
Diff. ($c = 60$)	0.01	0.0005	0.1430	0.0010	0.2670	0.0005	0.4875	1.0000
	0.05	0.0005	0.0915	0.0000	0.1685	0.0000	0.3555	0.9960
	0.10	0.0000	0.0670	0.0000	0.1225	0.0000	0.2775	0.9685
$y_t^*(c = 70)$	0.01	0.9975	0.6185	0.9990	0.5755	0.9995	0.3980	0.0000
	0.05	0.9995	0.7980	1.0000	0.7680	1.0000	0.5850	0.0000
	0.10	1.0000	0.8635	1.0000	0.8410	1.0000	0.6735	0.0010
Diff. ($c = 70$)	0.01	0.0005	0.1470	0.0010	0.2850	0.0005	0.5230	1.0000
	0.05	0.0005	0.0915	0.0000	0.1845	0.0000	0.3935	1.0000
	0.10	0.0000	0.0700	0.0000	0.1310	0.0000	0.3160	0.9990
Time (Sec.)		0.1257		0.1142		0.0971		

5.2 Model Framework

At each power level k , the KEP is a two-dimensional estimator, and we employ a two-dimensional CNN to address this supervised multi-class classification problem. Specifically, the network begins with an input layer that accepts the KEPs with a shape of $(99, 46, 1)$, where the final dimension corresponds to a single input channel. The first convolutional layer applies 32 filters with a kernel size of 3×3 , followed by a maxpooling layer with a pool size of 2×2 , and a dropout layer to reduce overfitting. The second convolutional block consists of 64 filters, again with a kernel size of 3×3 , followed by similar pooling and dropout operations. These convolutional blocks enable the model to learn increasingly abstract features while preserving the spatial integrity of the input. Following the convolutional layers, the output is flattened and passed through two fully connected layers, with 128 and 64 neurons, respectively, each followed by a ReLU activation. To enhance generalization and mitigate overfitting, dropout regularization is applied after the first dense layer. The final classification layer uses a softmax activation function to produce probability distributions over the target classes. The model is trained using categorical cross-entropy loss and the Adam optimizer with an initial

learning rate of 1e-3. To further improve convergence, a learning rate scheduler is employed to adaptively reduce the learning rate when the validation loss plateaus. The model architecture is shown in Figure 7. To classify the PGs, which are one-dimensional with respect to ω , we adapt the one-dimensional CNN model by setting the second dimension (corresponding to the number of expectile levels) equal to 1 (e.g. input size changed from (99, 46, 1) to (99, 1, 1); kernel size changed from 3×3 to 3×1 , and so on).

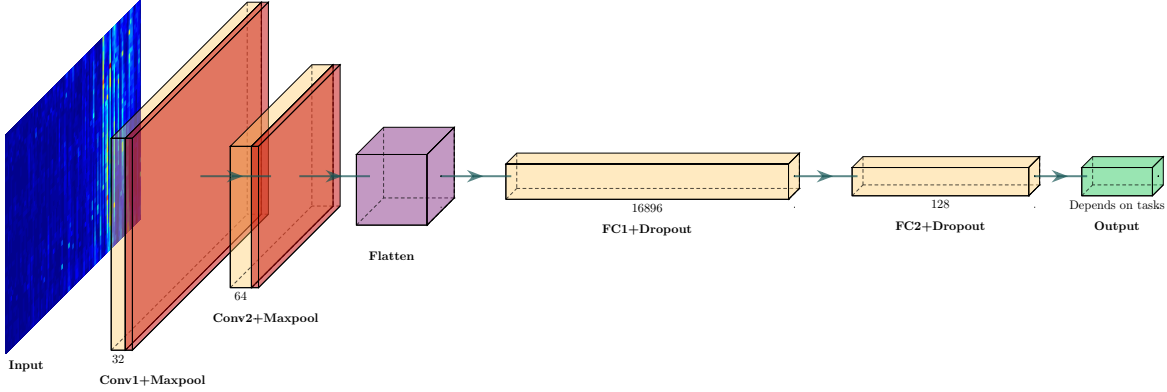


Figure 7: CNN model architecture.

5.3 Results and Comparisons

We randomly split the KEPs and PGs into training and validation sets, with proportions of 75% and 25%, respectively. The training repeats 10 times, each with a randomly constructed training–testing split and randomly initialized weights. Over 95% of the training runs converge within 30 epochs.

To evaluate the classifications, we consider the following metrics :

$$\text{Accuracy} = \frac{\sum_{i=1}^C (TP_i + TN_i)}{\sum_{i=1}^C (TP_i + TN_i + FP_i + FN_i)},$$

$$\text{Precision} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i},$$

$$\text{Recall} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i},$$

where C is the number of classes (set to 2 or 3 in different tasks). The classification outcomes are defined as follows:

- True Positive (TP_i): The number of instances correctly predicted as class i .
- False Positive (FP_i): The number of instances incorrectly predicted as class i but actually belonging to another class.

- False Negative (FN_i): The number of instances that belong to class i but were predicted as another class.
- True Negative (TN_i): The number of instances correctly predicted as not belonging to class i .

Table 2 shows the three metrics along with their standard deviations. The optimal value in each row is highlighted in bold. From the results, we can see that the classification based on the KEP achieves higher validation accuracy, precision, and recall rates than that of the PGs. As the task becomes more challenging, the performance gap between the KEP and the PG further increases, suggesting that the KEP provides more informative features for time series classification. Figure 8 illustrates two misclassifications of task A vs. E with $k = 1.1$. As can be seen, the KEP from group A exhibits multiple spectral peaks at $\omega \in (0, 0.08) \times 2\pi$, with the largest peaks around $\omega = 0.07 \times 2\pi$, while the KEP from group E exhibits a dominant peak at $\omega = 0.035 \times 2\pi$. In one misclassification shown in Figure 8(b), the KEP from group A has multiple spectral peaks at $\omega \in (0, 0.08) \times 2\pi$, but its largest peak is at $\omega = 0.035 \times 2\pi$, leading to misclassification. Similarly, in Figure 8(d), although the KEP has a large peak at $\omega = 0.07 \times 2\pi$, the presence of multiple small peaks at $\omega \in (0, 0.08) \times 2\pi$ results in misclassification.

Table 2: Classification results on validation set, along with the standard deviations. The best performance in each row is highlighted in bold.

Task	Metrics	KEP ($k = 1.1$)	KEP ($k = 1.5$)	KEP ($k = 1.9$)	PG
A vs. E (Easy)	Accuracy	0.9980 (0.0060)	0.9960 (0.0080)	0.9800 (0.0155)	0.9820 (0.0155)
	Precision	0.9979 (0.0062)	0.9857 (0.0085)	0.9801 (0.0156)	0.9819 (0.0156)
	Recall	0.9981 (0.0056)	0.9964 (0.0073)	0.9805 (0.0152)	0.9805 (0.0152)
ABCD vs. E (Intermediate)	Accuracy	0.9888 (0.0096)	0.9880 (0.0103)	0.9768 (0.0149)	0.9480 (0.0074)
	Precision	0.9932 (0.0059)	0.9927 (0.0064)	0.9859 (0.0090)	0.9680 (0.0042)
	Recall	0.9712 (0.0234)	0.9694 (0.0251)	0.9431 (0.0326)	0.8918 (0.0154)
A vs. D vs. E (Hard 1)	Accuracy	0.9773 (0.0134)	0.9780 (0.0123)	0.9773 (0.0120)	0.9053 (0.0071)
	Precision	0.9789 (0.0122)	0.9807 (0.0116)	0.9780 (0.0114)	0.9196 (0.0065)
	Recall	0.9781 (0.0116)	0.9802 (0.0116)	0.9779 (0.0101)	0.8992 (0.0072)
AB vs. CD vs. E (Hard 2)	Accuracy	0.9672 (0.0187)	0.9728 (0.0144)	0.9744 (0.0171)	0.9216 (0.0078)
	Precision	0.9742 (0.0146)	0.9748 (0.0159)	0.9791 (0.0138)	0.9330 (0.0085)
	Recall	0.9615 (0.0243)	0.9705 (0.0182)	0.9677 (0.0223)	0.9096 (0.0082)

6 Conclusion

We have proposed the KEP as a counterpart to the PG as a non-parametric tool for time series analysis. The KEP offers more information than the PG by examining serial dependence at different expectile levels, and the choice of power k makes the KEP a more flexible estimator compared to the QP and EP. We also established the asymptotic theory and investigated the relationship between the KEP and what we term KES. Simulations demonstrate the capability of the KEP in detecting hidden periodicities, while maintaining robustness against outliers. Finally, by leveraging the

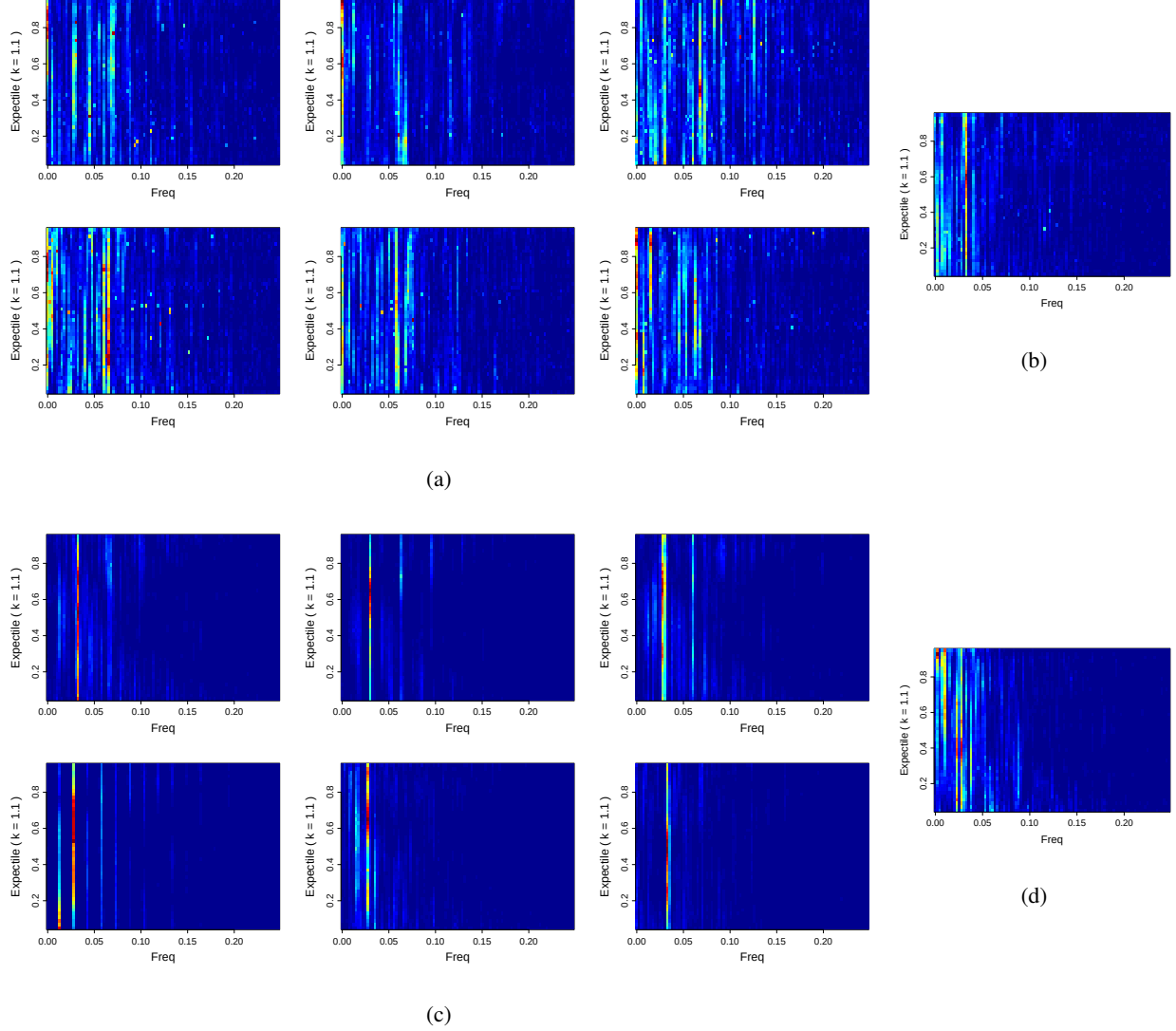


Figure 8: Two misclassifications. (a) and (c) the KEPs from group A and E, respectively. (b) The KEP from group A but incorrectly predicted as group E, and (d) the KEP from group E but incorrectly predicted as group A.

inherent two-dimensional nature of the KEP, we train a deep learning model to classify epilepsy EEG data. Notably, our approach outperforms the PG in terms of three different classification metrics.

However, the KEP is not free from limitations. First, when k approaches 1, the estimator exhibits strong robustness, whereas when k is close to 2, it becomes computationally and statistically more efficient. However, these are two sides of the same coin: as k approaches 1, the computational burden increases, and as k approaches 2, the robustness of the estimator decreases. This requires an additional step in choosing k . One can use method in Jiang et al. (2021), or guided by the specific objectives: whether the emphasis is placed on robustness or efficiency. Second, unlike the PG, which can be computed by the Fast Fourier Transform (FFT), the KEP is computationally more expensive using regression. A straightforward solution is to use fewer expectiles. In this paper, we selected a large number of expectiles uniformly

across $(0, 1)$. Researchers may choose to focus exclusively on a subset of expectiles with sufficient discriminative power (e.g., only high or low expectiles, since when α is close to 0.5, the behavior is analogous to the PG and LP). Furthermore, we utilize multi-thread parallelization to speed up computation using the packages `foreach`, `doParallel` in R.

In closing, further developments are needed. First, it is worthwhile to derive the joint asymptotic distribution of the KEP at multiple expectile levels to better understand the behavior of the KEP for expectile-frequency analysis. Second, one can develop another type of KEP using the same way to develop type II QP in Li (2012b). Finally, developing smoothing algorithms for the KEP would enhance its practical utility.

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Data and Code availability

The R code to reproduce the results, including all functions used in this paper, is available at <https://github.com/tianbochen1/K-th-Power-Expectile-Periodogram>.

Disclosure statement

The author reports there are no competing interests to declare.

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Appendix A

A.1 Lemma and Proof

In the proof, we drop the argument α and k for simplicity, as they are fixed numbers. Let the following assumptions be satisfied:

A1. The cumulative distribution function $F(\cdot)$ is Lipschitz continuous, i.e., there exists a constant $K > 0$ such that

$$|F(y) - F(y')| \leq K|y - y'| \text{ for all } y, y' \in \mathbb{R}.$$

A2. The regressor sequence $\{\mathbf{x}_{nt}\}$ is bounded.

A3. $E|y_t|^{k+2} < c_0$.

A4. $1.5 < k \leq 2$.

A5. There exists a positive definite matrix $\mathbf{D}_j$ for all j such that $\mathbf{D}_{jn} \rightarrow \mathbf{D}_j$ as $n \rightarrow \infty$, where

$$\mathbf{D}_{jn} := n^{-1} \sum_{t=1}^n \mathbf{x}_{jnt} \mathbf{x}_{jnt}^\top.$$

A6. There exists a positive definite matrix $\mathbf{V}_{jl}$ for all j, l such that $\mathbf{V}_{jln} \rightarrow \mathbf{V}_{jl}$ as $n \rightarrow \infty$, where

$$\mathbf{V}_{jln} := n^{-1} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}\{\dot{\rho}(y_t - \mu), \dot{\rho}(y_s - \mu)\} \mathbf{x}_{jnt} \mathbf{x}_{jns}^\top, \quad j, l = 1, \dots, q. \quad (18)$$

A7. A central limit theorem is valid for $\text{vec}[\boldsymbol{\zeta}_{jn}]_{j=1}^q$ for all j , i.e., $\text{vec}[\boldsymbol{\zeta}_{jn}]_{j=1}^q \xrightarrow{D} N(\mathbf{0}, [\mathbf{V}_{jl}]_{j,l=1}^q)$ as $n \rightarrow \infty$.

Lemma 1 (k -th Power Expectile Regression). If (A1) - (A6) are satisfied, then

$$\sqrt{n} \text{vec}[\hat{\boldsymbol{\beta}}_{jn} - \boldsymbol{\beta}_0]_{j=1}^q \xrightarrow{D} N(\mathbf{0}, [\boldsymbol{\Lambda}_j^{-1} \mathbf{V}_{jl} \boldsymbol{\Lambda}_l^{-1}]_{j,l=1}^q),$$

where

$$\hat{\boldsymbol{\beta}}_{jn} := \arg \min_{\boldsymbol{\beta}_j \in \mathbb{R}^p} \sum_{t=1}^n \rho(y_t - \mathbf{x}_{jnt}^\top \boldsymbol{\beta}_j),$$

$\boldsymbol{\beta}_0 := [\mu, 0, \dots, 0]^\top \in \mathbb{R}^p$, and

$$\boldsymbol{\Lambda}_j := \lim_{n \rightarrow \infty} \boldsymbol{\Lambda}_{jn} := \lim_{n \rightarrow \infty} \eta^{-1} \mathbf{D}_{jn} = \eta^{-1} \mathbf{D}_j.$$

Proof. The proof follows the general strategy used to establish similar results for the QP and EP, incorporating key results from Jiang et al. (2021). First, consider the case of $q = 1$, where we drop the subscripts j and l in the notation

for simplicity. Let $U_t := y_t - \mu = y_t - \mathbf{x}_{nt}^T \boldsymbol{\beta}_0$ and $c_{nt}(\boldsymbol{\delta}) := n^{-1/2} \mathbf{x}_{nt}^T \boldsymbol{\delta}$ with $\boldsymbol{\delta} \in \mathbb{R}^p$, consider the random function

$$Z_n(\boldsymbol{\delta}) := \sum_{t=1}^n \{\rho(U_t - c_{nt}(\boldsymbol{\delta})) - \rho(U_t)\}.$$

Note that, by reparameterizing $\boldsymbol{\delta} := \sqrt{n}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)$ as a function of $\boldsymbol{\beta} \in \mathbb{R}^p$, one can write

$$Z_n(\boldsymbol{\delta}) = \sum_{t=1}^n \{\rho(y_t - \mathbf{x}_{nt}^T \boldsymbol{\beta}) - \rho(U_t)\}.$$

Therefore, the minimizer of $Z_n(\boldsymbol{\delta})$ over $\boldsymbol{\delta} \in \mathbb{R}^p$ takes the form

$$\hat{\boldsymbol{\delta}}_n := \arg \min_{\boldsymbol{\delta} \in \mathbb{R}^p} Z_n(\boldsymbol{\delta}) = \sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0).$$

We aim to prove

$$\hat{\boldsymbol{\delta}}_n = \boldsymbol{\Lambda}_n^{-1} \boldsymbol{\zeta}_n + o_P(1). \quad (19)$$

In order to arrive at (19), we first would like to prove that

$$Z_n(\boldsymbol{\delta}) = \tilde{Z}_n(\boldsymbol{\delta}) + o_P(1), \quad (20)$$

for any fixed $\boldsymbol{\delta}$, where $\tilde{Z}_n(\boldsymbol{\delta}) := -\boldsymbol{\delta}^T \boldsymbol{\zeta}_n + (1/2) \boldsymbol{\delta}^T \boldsymbol{\Lambda}_n \boldsymbol{\delta}$. If (20) holds, then, due to the convexity of $Z_n(\boldsymbol{\delta})$ and $\tilde{Z}_n(\boldsymbol{\delta})$ as functions of $\boldsymbol{\delta}$, one can follow the arguments in Li (2008) to show that $\hat{\boldsymbol{\delta}}_n$ as the minimizer of $Z_n(\boldsymbol{\delta})$ is $o_P(1)$ away from the minimizer of $\tilde{Z}_n(\boldsymbol{\delta})$, which equals $\boldsymbol{\Lambda}_n^{-1} \boldsymbol{\zeta}_n$. This completes the proof of (19).

To prove (20), using the Taylor formula, we have

$$\begin{aligned} \rho(U_t - c_{nt}) - \rho(U_t) &= -\dot{\rho}(U_t) c_{nt} + \frac{1}{2} c_{nt}^2 \ddot{\rho}(U^*) \\ &:= -\dot{\rho}(U_t) c_{nt} + R_{nt}, \end{aligned}$$

where U^* is between U_t and c_{nt} , and $\ddot{\rho}(u)$ is defined in (10). Then, by taking summation over t , we obtain

$$Z_n(\boldsymbol{\delta}) = -\boldsymbol{\delta}^T \boldsymbol{\zeta}_n + \sum_{t=1}^n R_{nt},$$

Here, we drop the argument $\boldsymbol{\delta}$ from R_{nt} and c_{nt} for simplicity of notation as $\boldsymbol{\delta}$ is fixed in the remainder of the proof.

By Theorem 4 in Jiang et al. (2021) under assumptions (A1), (A3), and (A4), it follows that

$$\mathbb{E} \left\{ \sum_{t=1}^n R_{nt} \right\} = \frac{1}{2} \mathbb{E} \{ \ddot{\rho}(y_t - \mu) \} c_{nt}^2 + o(c_{nt}^2),$$

and

$$\text{Var} \left\{ \sum_{t=1}^n R_{nt} \right\} = o(c_{nt}^2).$$

Under assumption (A2), $c_{nt}^2 = o(n^{-1})$. By substituting $c_{nt} = n^{-1/2} \mathbf{x}_{nt}^T \boldsymbol{\delta}$, we have

$$\mathbb{E} \left\{ \sum_{t=1}^n R_{nt} \right\} = \frac{1}{2} \boldsymbol{\delta}^T \boldsymbol{\Lambda}_n \boldsymbol{\delta} + o(1), \quad (21)$$

and

$$\text{Var} \left\{ \sum_{t=1}^n R_{nt} \right\} = o(1). \quad (22)$$

Combining (21) and (22), we obtain

$$\sum_{t=1}^n R_{nt} = \frac{1}{2} \boldsymbol{\delta}^T \boldsymbol{\Lambda}_n \boldsymbol{\delta} + o_P(1), \quad (23)$$

which establishes (20). Therefore,

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0) = \boldsymbol{\Lambda}_n^{-1} \boldsymbol{\zeta}_n + o_P(1). \quad (24)$$

Next, we aim to establish the asymptotic normality of the KER solution. Because μ satisfies (1), it follows that

$$\mathbb{E}\{\boldsymbol{\zeta}_n\} = n^{-1/2} \sum_{t=1}^n \mathbb{E}\{\dot{\rho}(y_t - \mu)\} \mathbf{x}_{nt} = \mathbf{0}.$$

Moreover, we have

$$\mathbf{V}_n := \text{Cov}\{\boldsymbol{\zeta}_n\} = n^{-1} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}\{\dot{\rho}(y_t - \mu), \dot{\rho}(y_s - \mu)\} \mathbf{x}_{nt} \mathbf{x}_{ns}^\top.$$

Then, under assumptions (A5)-(A7) and (24), we obtain

$$\boldsymbol{\Lambda}_n^{-1} \boldsymbol{\zeta}_n \xrightarrow{D} N(\mathbf{0}, \boldsymbol{\Lambda}^{-1} \mathbf{V} \boldsymbol{\Lambda}^{-1}).$$

For the general case of $q > 1$, we reintroduce the subscript j in the notation. By replacing $\mathbf{x}_{nt}$ with $\mathbf{x}_{jnt}$, we obtain $Z_{jn}(\boldsymbol{\delta}_j)$ and

$$Z_n^*(\boldsymbol{\delta}^*) = \sum_{j=1}^q Z_{jn}(\boldsymbol{\delta}_j),$$

where $\boldsymbol{\delta}^* := \text{vec}[\boldsymbol{\delta}_j]_{j=1}^q$, and $\boldsymbol{\delta}_j := \sqrt{n}(\boldsymbol{\beta}_j - \boldsymbol{\beta}_0)$. By a similar argument, it can be shown that the minimizer of $Z_n^*(\boldsymbol{\delta}^*)$, which can be expressed as $\hat{\boldsymbol{\delta}}_n^* := \text{vec}[\hat{\boldsymbol{\delta}}_{jn}]_{j=1}^q$, is $o_P(1)$ away from $\tilde{\boldsymbol{\delta}}_n^* := \text{vec}[\tilde{\boldsymbol{\delta}}_{jn}]_{j=1}^q := \text{vec}[\boldsymbol{\Lambda}_{jn}^{-1} \boldsymbol{\zeta}_{jn}]_{j=1}^q$, where $\tilde{\boldsymbol{\delta}}_{jn}$ is the minimizer of $Z_{jn}(\boldsymbol{\delta}_j)$. Therefore, $\hat{\boldsymbol{\delta}}_n^*$ has the same asymptotic distribution as $\tilde{\boldsymbol{\delta}}_n^*$. Observe that $\mathbb{E}\{\boldsymbol{\zeta}_{jn}\} = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\zeta}_{jn}, \boldsymbol{\zeta}_{ln}) = \mathbf{V}_{jln}$. Therefore, $\hat{\boldsymbol{\delta}}_n^* \xrightarrow{D} N(\mathbf{0}, [\boldsymbol{\Lambda}_j^{-1} \mathbf{V}_{jl} \boldsymbol{\Lambda}_l^{-1}]_{j,l=1}^q)$, and **Lemma 1** is proved.

A.2 Proof of Theorem 1

For fixed $q > 1$ and $0 < \lambda_1 < \dots < \lambda_q < \pi$, let $\omega_{\nu_1}, \dots, \omega_{\nu_q}$ be Fourier frequencies satisfying $\omega_{\nu_j} \rightarrow \lambda_j$ as $n \rightarrow \infty$ for $j = 1, \dots, q$. The trigonometric regressor $\mathbf{x}_{jnt} := [1, \cos(\omega_{\nu_j} t), \sin(\omega_{\nu_j} t)]^\top$ is bounded, so (A2) is satisfied. It is easy to verify that as $n \rightarrow \infty$,

$$\mathbf{D}_{jn} = n^{-1} \sum_{t=1}^n \mathbf{x}_{jt}(\omega_{\nu_j}) \mathbf{x}_{jt}^\top(\omega_{\nu_j}) \rightarrow \mathbf{D}_j = \text{diag}\{1, 1/2, 1/2\},$$

so (A5) is satisfied. It follows from (18) that, when $j = k$,

$$\mathbf{V}_{jjn} = \sum_{|\tau| < n} \gamma(\tau) \left\{ n^{-1} \sum_{t=\max(1, 1+\tau)}^{\min(n, n+\tau)} \mathbf{x}_{jt}(\omega_{\nu_j}) \mathbf{x}_{j, t-\tau}^\top(\omega_{\nu_j}) \right\}.$$

For fixed τ , we have

$$n^{-1} \sum_{t=\max(1, 1+\tau)}^{\min(n, n+\tau)} \mathbf{x}_{jt}(\omega_{\nu_j}) \mathbf{x}_{j, t-\tau}^\top(\omega_{\nu_j}) \rightarrow \text{diag}\{1, (1/2)\mathbf{S}(\lambda_j)\},$$

where

$$\mathbf{S}(\lambda_j) := \begin{bmatrix} \cos(\lambda_j \tau) & -\sin(\lambda_j \tau) \\ \sin(\lambda_j \tau) & \cos(\lambda_j \tau) \end{bmatrix}.$$

Then,

$$\begin{aligned}
\mathbf{V}_{jj} &:= \lim_{n \rightarrow \infty} \mathbf{V}_{jjn} \\
&= \sum_{\tau=-\infty}^{\infty} \gamma(\tau) \text{diag}\{1, (1/2)\mathbf{S}(\lambda_j)\} \\
&= \text{diag}\{h(0), (1/2)h(\lambda_j), (1/2)h(\lambda_j)\}.
\end{aligned} \tag{25}$$

When $j \neq l$,

$$\mathbf{V}_{jl} := \lim_{n \rightarrow \infty} \mathbf{V}_{jln} = \mathbf{0}. \tag{26}$$

Based on (25) and (26), we have

$$\mathbf{V}_{jl} = \delta_{j-l} \text{diag}\{h(0), (1/2)h(\lambda_j), (1/2)h(\lambda_j)\},$$

where δ_s denotes the Kronecker delta function, and

$$[\mathbf{\Lambda}_j^{-1} \mathbf{V}_{jl} \mathbf{\Lambda}_l^{-1}]_{j,l=1}^q = \text{diag}\{g(0), 2g(\lambda_1), 2g(\lambda_1), \dots, g(0), 2g(\lambda_q), 2g(\lambda_q)\}.$$

Hence, (A6) is satisfied. Moreover, (C1), (C3), (C4), and (C5) repeat (A1), (A3), (A4), and (A7), respectively. Finally, all assumptions (A1)-(A7) are satisfied. Therefore, by **Lemma 1**,

$$\sqrt{n} \text{vec}[\hat{\beta}_{jn}(\omega_{\nu_j}) - \beta_0]_{j=1}^q \xrightarrow{D} N(\mathbf{0}, \text{diag}\{g(0), 2g(\lambda_1), 2g(\lambda_1), \dots, g(0), 2g(\lambda_q), 2g(\lambda_q)\})$$

Further, because $\hat{\beta}_n(\omega_{\nu_j}) = [\hat{\beta}_1(\omega_{\nu_j}), \hat{\beta}_2(\omega_{\nu_j}), \hat{\beta}_3(\omega_{\nu_j})]^\top$, we have

$$\sqrt{n}[\hat{\beta}_2(\omega_{\nu_j}), \hat{\beta}_3(\omega_{\nu_j})]^\top / \sqrt{2g(\lambda_j)} \xrightarrow{D} N(\mathbf{0}, \mathbf{I}),$$

and hence

$$\text{KEP}_n(\omega_{\nu_j})/g(\lambda_j) = \frac{n}{4} \{\hat{\beta}_2^2(\omega_{\nu_j}) + \hat{\beta}_3^2(\omega_{\nu_j})\}/g(\lambda_j) \sim \frac{1}{2}(\xi_1^2 + \xi_2^2) = \chi_2^2,$$

where ξ_1 and ξ_2 are i.i.d. $N(0, 1)$ random variables. This implies that $\text{KEP}_n(\omega_{\nu_j})$ is asymptotically independent with distribution $g(\lambda_j)(1/2)\chi_{2,j}^2$ for each j , and we obtain (11). Finally, **Theorem 1** is proved.